

CSCE 669 COMPUTATIONAL OPTIMIZATION · MODULE 10 · LECTURE NOTES

Approximation and Relaxation

Provably close, provably fast.

MODULE OUTCOMES

- Define an approximation ratio and a PTAS.
- Derive a guarantee from an LP relaxation and rounding.
- Apply randomised rounding and its analysis.
- Explain semidefinite relaxation and the Goemans–Williamson result.
- Explain hardness of approximation.

1 The framework

THE GUARANTEE IS AGAINST A BOUND, NOT AGAINST THE OPTIMUM

You cannot compare your algorithm's output to the optimum, because computing the optimum is exactly the intractable problem you are avoiding. This is the obstacle that makes approximation analysis look impossible at first.

So compare it to a *bound* on the optimum instead — almost always the value of a relaxation, which is a lower bound for a minimisation problem by weak duality (Module 04 §1).

If your solution's cost is within a factor α of the bound, it is within α of the optimum, because the optimum lies between the bound and your solution. **The optimum never has to be computed; it only has to be sandwiched.**

That is the entire mechanism of approximation analysis. Every ratio in this module is proved against a relaxation, which is why Module 04's duality had to come first — and why this module, Module 04, and Module 09's branch and bound are all applications of one idea.

Class	What it means	Example
FPTAS	A $(1+\epsilon)$ -approximation in time polynomial in both n and $1/\epsilon$. The best you can hope for short of exact.	Knapsack, via dynamic programming on scaled profits.
PTAS	A $(1+\epsilon)$ -approximation in time polynomial in n for each fixed ϵ — the dependence on $1/\epsilon$ may be terrible.	Euclidean TSP (Arora, Mitchell), where the exponent depends on $1/\epsilon$.

Class	What it means	Example
Constant factor	Within a fixed ratio α .	Vertex cover (2), MAX-CUT (1.138 by Goemans–Williamson), metric TSP (3/2).
Logarithmic	Within $O(\log n)$.	Set cover — and the logarithm is provably optimal (§4).
No constant factor	No constant-factor approximation exists unless $P = NP$.	General (non-metric) TSP, maximum clique.

These classes are provably distinct under standard complexity assumptions, so the hierarchy reflects genuine structure rather than current ignorance — which is a strong and somewhat surprising thing to be able to say about approximation.

2 LP rounding

VERTEX COVER: minimum-weight set of vertices touching every edge.

INTEGER PROGRAM

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minimise  sum w_v x_v
s.t.      x_u + x_v >= 1   for every edge (u,v)
          x_v in {0,1}

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RELAX x_v in $[0,1]$, solve the LP, then ROUND:
 take v if $x_v \geq 1/2$

FEASIBLE: every edge has $x_u + x_v \geq 1$, so at least one endpoint has $x \geq 1/2$ and is taken. Every edge is covered.

COST: rounding at most DOUBLES each variable, so
 $ALG \leq 2 * LP \leq 2 * OPT$
 since $LP \leq OPT$ for a relaxed minimisation.

A 2-APPROXIMATION, PROVED, IN TEN LINES.

And the analysis shows where the 2 comes from: the threshold.
 The INTEGRALITY GAP of this formulation IS 2 (take a triangle:
 LP gives 1/2 everywhere for cost 3/2; the true optimum is 2).
 So no better rounding of THIS relaxation exists.

The integrality gap bounds what *any* rounding of that relaxation can achieve. To beat the factor you need a stronger formulation — more constraints, or a semidefinite relaxation (§3) — **not a cleverer rounding rule. That observation connects this module directly to Module 09's formulation strength:** the same quantity governs branch-and-bound tree size and approximation ratio, which is not a coincidence.

RANDOMISED ROUNDING, AND WHY IT ANALYSES WELL

Treat each fractional value x_v as a *probability* and include v independently with that probability. The idea is almost too simple to be a technique.

The expected cost is exactly the LP value, by linearity of expectation — and linearity holds regardless of dependence, which is why the cost analysis is a single line and requires no independence assumption at all.

Feasibility holds only in expectation, so a single round may leave constraints violated. Repeat the rounding, or round conditionally, and use concentration bounds (Chernoff; CSCE 658) to control the probability of failure.

Set cover's $O(\log n)$ guarantee comes from exactly this: round repeatedly, and after $O(\log n)$ independent rounds every element is covered with high probability, at $O(\log n)$ times the LP cost. **Which matches the hardness bound of §4** — the technique is optimal for that problem.

3 Semidefinite relaxation

GOEMANS–WILLIAMSON: RELAX TO VECTORS, ROUND BY A RANDOM HYPERPLANE

MAX-CUT asks for a partition of the vertices maximising the total weight of edges crossing it. Assign each vertex a value ± 1 and maximise $\sum_{ij} w_{ij}(1 - x_i x_j)/2$ — which is non-convex because of the discrete constraint.

Relax it by assigning each vertex a unit vector in high-dimensional space instead of a scalar, and maximising the same expression with the product replaced by an inner product. **That is a semidefinite program** — convex (Module 01 §2) and solvable in polynomial time.

Then round with a random hyperplane through the origin: the vertices whose vectors fall on each side form the two sides of the cut. A uniformly random direction, and nothing more.

The probability that an edge is cut is θ/π , where θ is the angle between its endpoints' vectors — the fraction of hyperplane orientations that separate them. **The ratio of θ/π to the SDP's contribution for that edge is at least 0.878 for every angle,** which is a one-variable calculus exercise. **A 0.878-approximation, derived from geometry,** and it is one of the most elegant results in algorithms.

AND 0.878 IS PROBABLY OPTIMAL

Under the Unique Games Conjecture, no polynomial-time algorithm achieves a better ratio than 0.878 for MAX-CUT.

So the Goemans–Williamson constant — which looks entirely like an artifact of their particular analysis, being the minimum of a transcendental ratio — is the true boundary of what is achievable.

That is a remarkable convergence. An algorithmic result and a hardness result, developed independently by different methods, meeting on an arbitrary-looking number — which is reasonable evidence that the number is describing something real about the problem rather than about the technique.

The Unique Games Conjecture is unproved and widely believed, and a great deal of modern approximation hardness is conditional on it. That conditionality should be stated when citing these results — 'optimal unless UGC fails' is the honest form, and it is a weaker claim than 'optimal unless $P = NP$ '.

4 Hardness of approximation

- **Set cover:** no approximation better than $(1 - o(1)) \cdot \ln n$ unless $P = NP$. **And the simple greedy algorithm achieves exactly that — the most obvious method is provably optimal,** which is both satisfying and a useful warning against assuming sophistication helps.
- **Maximum clique:** no $n^{1-\epsilon}$ approximation unless $P = NP$. **Essentially inapproximable** — you cannot do meaningfully better than returning a single vertex.
- **General TSP:** no constant-factor approximation at all — **and metric TSP has a 3/2-approximation** (Christofides, 1976, improved only in 2021). **The triangle inequality is what makes the difference,**

which is a good illustration that a problem's approximability depends on structure rather than on its exact-solution complexity.

- **Vertex cover:** no better than 2 under UGC. **So the ten-line algorithm of §2 is optimal**, and the decades of effort spent trying to beat it were spent against a wall.
- **The PCP theorem is what makes these results provable** — a characterisation of NP in terms of probabilistically checkable proofs, which converts approximation hardness into a gap-preserving reduction. **It is one of the deepest results in complexity theory** (CSCE 637), and its practical consequence here is that **'nobody has found a better algorithm' can be upgraded to 'no better algorithm exists'**.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Williamson & Shmoys — The Design of Approximation Algorithms (free PDF)	Free in full, and the reference for this entire module. The LP rounding and randomised rounding chapters cover §2 completely.
Goemans & Williamson — Improved Approximation Algorithms for Maximum Cut (free)	The §3 result. The geometric argument is followable and worth working through.
Vazirani — Approximation Algorithms (free PDF)	Free, and complementary — stronger on the primal-dual method and on the combinatorial algorithms.
Khot — On the Unique Games Conjecture (free survey)	The §3 and §4 conditionality , explained by the person who proposed the conjecture.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement the vertex cover 2-approximation** and verify the ratio empirically against exact solutions on small instances.
- 2 **Construct the triangle instance** showing the integrality gap is exactly 2.
- 3 Implement greedy set cover and measure its ratio against the LP bound on random instances.
- 4 **Implement randomised rounding for set cover** and measure how many rounds are needed for feasibility.
- 5 **Solve a small MAX-CUT by semidefinite programming** with CVXPY and round with a random hyperplane.
- 6 **Repeat the rounding many times** and plot the distribution of cut values against the SDP bound.

- 7 Compare the SDP relaxation's bound against the LP relaxation's on the same instances.
- 8 **Measure the achieved ratio** and compare against 0.878.
- 9 Take a problem from your own work and **find or derive an approximation guarantee** for a simple algorithm on it.
- 10 **Compute the integrality gap** of a formulation you wrote in Module 09.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 How is an approximation guarantee proved without computing the optimum?
- 2 Name five approximability classes and an example of each.
- 3 Derive the vertex cover 2-approximation and say where the 2 comes from.
- 4 What does the integrality gap bound?
- 5 Why does randomised rounding analyse cleanly?
- 6 Describe the Goemans–Williamson relaxation and rounding.
- 7 What is the status of the 0.878 bound?
- 8 Give four inapproximability results and say which algorithms are already optimal.